Toulouse School of Economics, 2011-2012
Macroeconomics – Christian Hellwig, Franck Portier & Gilles Saint Paul

Homework 2 (FP)
Version 2 (changes in red)

Problem I – A Simplified Real-Business-Cycle Model with Additive Technology Shocks

Consider an economy consisting of a constant population (of mass one) of infinitely-lived individuals. The representative individual maximizes the expected value of $\sum_{t=0}^{\infty} u(C_t)/(1+\rho)^t$, $\rho > 0$. The instantaneous utility function, $u(C_t)$, is $u(C_t) = C_t - \theta C_t^2$, $\theta > 0$. Assume that C is always in the range where $u'(C)$ is positive.

Output is linear in capital, plus an additive disturbance: $Y_t = AK_t + e_t$. There is no depreciation; thus $K_{t+1} = K_t + Y_t - C_t$, and the interest rate is A. Assume $A = \rho$. Finally, the disturbance follows a first-order autoregressive process: $e_t = \phi e_{t-1} + \varepsilon_t$, where $-1 < \phi < 1$ and where the ε_t's are zero mean i.i.d. shocks.

1 – Find the first-order condition (Euler equation) relating C_t and expectations of C_{t+1}.

2 – Guess that consumption takes the form $C_t = \alpha + \beta K_t + \gamma e_t$. Given this guess, what is K_{t+1} as a function of K_t and e_t?

3 – What values must the parameters α, β, and γ have for the first-order condition in part (1) to be satisfied for all values of K_t and e_t?

4 – What are the effects of a one-time shock to ε on the paths of Y, K, and C?

Problem II – Recursive Formulation of the Simple RBC Model

Consider a competitive complete market economy. The infinitely lived representative agent maximizes the expectation of $\sum_{t=0}^{\infty} e^{-\rho t} u(C_t, 1-L_t)$, $\rho > 0$. Instantaneous utility is given by $u(C_t, 1-L_t) = \log c_t + b \log(1-L_t)$. The aggregate production funcion is $Y_t = K_t^{\alpha}(A_t L_t)^{1-\alpha}$. It is assumed that capital fully depreciates within one period. A_t is a random variable whose stochastic process is : $A_t = \rho_A A_{t-1} + \varepsilon_t$, with ε iid with zero mean and $0 \le \rho < 1$.

The *value function* $V(K_t, A_t)$ is defined as the expected discounted value in t of the sum of current and future periods utilities, as a function of current capital stock and technology.

1 – Explain why $V(\bullet)$ must satisfy

$$V(K_t, A_t) = \max_{C_t, L_t} \left(u(C_t, 1-L_t) + e^{-\rho} E_t[V(K_{t+1}, A_{t+1}]\right)$$

This condition is known as the Bellman equation. Given the log-linear structure of the model, let's guess that V is of the following parametric form: $V(K, A) = \beta_0 + \beta_K \log K + \beta_A \log A$, where the β are unknown parameters that we need to determine.

2 – Show that one can obtain the following expression for the Bellman equation:

$$\begin{aligned} V(K_t, A_t) &= \max_{C_t, L_t} \Big\{ u(C_t, 1-L_t) \\ &\quad + e^{-\rho}(\beta_0 + \beta_K \log(Y_t - C_t) + \beta_A \rho_A \log A_t) \Big\} \end{aligned}$$

3 – Find the first order optimality condition for C_t. Show that C_t/Y_t does not depend on K_t and A_t.

4 – Find the first order optimality condition for L_t. Show that L_t does not depend on K_t and A_t.

5 – Using the results of the previous questions, show that the value function can be written $V(K, A) = \beta_0' + \beta_K' K + \beta_A' A$

6 – Find β_K and β_A such that $\beta_K = \beta_K'$ et $\beta_A = \beta_A'$? (we don't compute β_0)

7 – What are the values for C/Y and L?

Consider a model economy populated with a representative household and a representative firm. The firm has a Cobb-Douglas technology:

$$Y_t = Z_t K_t^{\gamma} N_t^{1-\gamma} \tag{1}$$

where K_t is capital, N_t labor input, and Z_t a stochastic technological shock. All profits of the firm are distributed to the household. Capital evolves according to the log linear relation

$$K_{t+1} = A K_t^{1-\delta} I_t^{\delta} \qquad 0 < \delta \leq 1 \tag{2}$$

where δ is the rate of depreciation and where I_t is investment in period t.

The representative household works N_t, consumes C_t and invests I_t in period t. Preferences are given by:

$$U = E_0 \sum_{t=0}^{\infty} \beta^t \left[\log C_t - V(N_t)\right] \tag{3}$$

where V is a convex function. Capital is accumulated by the household and rented to the firm.

Let κ denote the real rental rate of capital, P the price of the final good and W the nominal wage.

We first assume that $\delta = 1$ (full depreciation).

1 – Write down the budget constraint of the household and the profit function of the firm

2 – Derive FOCs of the utility and profit maximization

3 – Define a competitive equilibrium of this economy

4 – Solve the model and show that N_t is constant along an equilibrium path.

5 – Derive a $AR(1)$ process for log of output. Draw the Impulse Response Function of y to a unit shock to $\log Z = z$, assuming that z is *iid*. What are the determinants of the size and persistence of this IRF?

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We assume now that $\delta \in]0, 1[$.

6 – What is the economic meaning of equation (2)?

7 – We still denote κ_t the real return in t on investment I_{t-1}:

$$\kappa_t = \frac{\partial Y_t}{\partial I_{t-1}} \tag{4}$$

and note that the real price, in terms of current output, of capital to be transmitted to the next period is not one as when $\delta = 1$. We denote it as q_t (like Tobin's q), and it is equal to:

$$q_t = \frac{\partial K_{t+1}/\partial K_t}{\partial K_{t+1}/\partial I_t} \tag{5}$$

8 – Comment the new budget constraint of the household

$$C_t + I_t = \frac{W_t}{P_t} N_t + \kappa_t (I_{t-1} + q_{t-1} K_{t-1}) \tag{6}$$

9 – Solve for the competitive equilibrium of this economy and compute $\frac{I_t}{C_t}$.

10 – Compute the constant level of N.

11 – Derive the new process for log of output. Draw output typical IRF and discuss of its shape as a function of parameters.

Problem IV – Technological shocks, Preference shocks and the endogeneity of TFP

We consider here a variation of the simple analytical RBC model. We study a model economy $\mathcal{A}$ populated with a representative household and a representative firm. The firm has a Cobb-Douglas technology:

$$Y_t = Z_t K_t^{\gamma} N_t^{1-\gamma} \tag{7}$$

where K_t is capital, N_t labor input, and Z_t the stochastic Total Factor Productivity (TFP). All profits of the firm are distributed to the household. Capital evolves according to

$$K_{t+1} = I_t \tag{8}$$

where I_t is investment in period t.

The representative household works N_t and consumes C_t. Preferences are given by

$$U = E_0 \sum_{t=0}^{\infty} \beta^t \left[\log C_t - \chi_t N_t\right] \tag{9}$$

where χ_t is a preference shock. Capital is accumulated by the household and rented to the firm.

Let κ denote the real rental rate of capital, P the price of the final good and W the nominal wage.

1 – Write down the budget constraint of the household and the profit function of the firm

2 – Derive FOCs of the utility and profit maximization

3 – Define a competitive equilibrium of this economy

4 – Solve the model and show that the equilibrium process of output is $y_t = z_t + \gamma y_{t-1} - (1-\gamma)\chi_t$ (1) (dropping constants and with the notation $x = \log X$)

5 – Assume $y_{-1} = 0$, $z_t = 0\ \forall t$, $\chi_t = 0\ \forall t$, except $\chi_0 = 1$. Draw the time path on χ, z and y. Explain why y is persistent

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We now consider an economy $\mathcal{B}$, in which the TFP is not exogenous at the aggregate level, but given by $Z_t = \overline{Y}_t^{\theta} X_t$, where X is the exogenous part of TFP and $\overline{Y}_t^{\theta}$ act as an externality. More precisely, $\overline{Y}$ is taken as given by firms and households, but one has at the competitive equilibrium $\overline{Y} = Y$

6 – What is the economic interpretation of this externality?

7 – Solve for the competitive equilibrium and give the equilibrium process of output (again in logs). Comment

8 – Assume that an economist observes the economy $\mathcal{B}$, with externality, but thinks that he is observing economy $\mathcal{A}$, and is therefore using equation (1) to understand the data. Draw the response of observed TFP z_t, of y and χ if $y_{-1} = 0$, $x_t = 0\ \forall t$, $\chi_t = 0\ \forall t$, except $\chi_0 = 1$.

9 – How to interpret the positive correlation between observed TFP and output? Could such an economist believe (wrongly) that technological shocks are driving part of the response of the economy? Discuss.

Problem V - The analytical RBC Model and the Real Wage - Employment Correlation

Let the model economy be populated with a representative household and a representative firm. The firm has a Cobb-Douglas technology:

$$Y_t = e^{z_t} K_t^{\gamma} N_t^{1-\gamma} \tag{10}$$

where K_t is capital, N_t labor input, and z_t a stochastic technological shift. All profits of the firm are distributed to the household. Capital evolves according to

$$K_{t+1} = I_t \tag{11}$$

where I_t is investment in period t.

The representative household works N_t and consumes C_t. Preferences are given by

$$U = E_0 \sum_{t=0}^{\infty} \beta^t \left[\log C_t - e^{\chi_t} N_t\right] \tag{12}$$

where χ_t is a preference shock. Capital is accumulated by the household and rented to the firm. Let R_t denote the real rental rate of capital and W_t the real wage. The final good is chosen as the numéraire. It is assumed that χ and z are *i.i.d.* with respective variance σ_χ^2 and σ_z^2.

1 – Write the representative household problem and derive the FOCs.

2 – Write the representative firm problem and derive the FOCs.

3 – Define a competitive equilibrium of this economy

4 – Solve the model and show that the equilibrium process of output is $y_t = z_t + \gamma y_{t-1} - (1-\gamma)\chi_t$ (dropping constants and with the notation $x = \log X$)

5 – Derive the solution for the (log of the) real wage ω_t and for employment n_t (again dropping constants).

6 – Compute and draw the IRF of y, ω and n to a technological and preference shock. Discuss.

7 – Compute the correlation between ω_t and n_t. What do you know about the level of this correlation in the data. Discuss.